

$$\text{If } \vec{v}_1 = (x, y, z) \in \mathbb{R}^3 \\ \vec{v}_2 = (a, b, c) \in \mathbb{R}^3$$

$\langle \cdot, \cdot \rangle := \text{scalar product}$

$$\langle \vec{v}_1, \vec{v}_2 \rangle = ax + by + cz.$$

Properties of the scalar product:

$$1) \langle \vec{v}_1, \vec{v}_2 \rangle = \langle \vec{v}_2, \vec{v}_1 \rangle \text{ (comm.)}$$

$$2) \langle \vec{v}_1 + \vec{v}_2, \vec{v}_3 \rangle = \langle \vec{v}_1, \vec{v}_3 \rangle + \langle \vec{v}_2, \vec{v}_3 \rangle \text{ (distrib.)}$$

$$3) \langle \vec{v}_1, \vec{v}_2 \rangle = 0 \Leftrightarrow \vec{v}_1 \perp \vec{v}_2.$$

$$4) \|\vec{v}_1\| := \text{length } \vec{v}_1 = \sqrt{x^2 + y^2 + z^2} = \sqrt{\langle \vec{v}_1, \vec{v}_1 \rangle}$$

$$\Rightarrow \langle \vec{v}_1, \vec{v}_1 \rangle = \|\vec{v}_1\|^2.$$

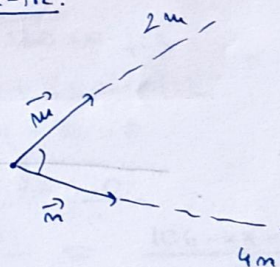
$$5) \langle \vec{v}_1, \vec{v}_2 \rangle = \|\vec{v}_1\| \cdot \|\vec{v}_2\| \cdot \cos(\angle \vec{v}_1, \vec{v}_2)$$

when $= 0 \Rightarrow \perp$

3.1. Let u, m unit vectors s.t. $\angle(u, m) = 120^\circ \Rightarrow \|\vec{u}\| = \|\vec{m}\| = 1$

Det. the angle between $a = 2\vec{u} + 4\vec{m}$ and $b = \vec{u} - \vec{m}$.

$$\begin{aligned} \langle \vec{u}, \vec{m} \rangle &= \|\vec{u}\| \cdot \|\vec{m}\| \cdot \cos(\angle \vec{u}, \vec{m}) \\ &= 1 \cdot 1 \cdot \cos 120^\circ \\ &= \cos(180^\circ - 60^\circ) \\ &= -\frac{1}{2}. \end{aligned}$$



$$\langle \vec{a}, \vec{b} \rangle = \|\vec{a}\| \cdot \|\vec{b}\| \cdot \cos(\angle \vec{a}, \vec{b}) = \dots$$

$$\begin{aligned} \|\vec{a}\| &= \sqrt{\langle \vec{a}, \vec{a} \rangle} = \sqrt{\langle 2\vec{u} + 4\vec{m}, 2\vec{u} + 4\vec{m} \rangle} = \sqrt{2\langle \vec{u}, 2\vec{u} \rangle + 2\langle \vec{u}, 4\vec{m} \rangle + \langle 4\vec{m}, 4\vec{m} \rangle} \\ &= \sqrt{112\|\vec{u}\|^2 + 2 \cdot \|\vec{u}\| \cdot \|\vec{m}\| \cdot \cos(\angle \vec{u}, \vec{m}) + 16\|\vec{m}\|^2} \\ &= \sqrt{4 \cdot 1 + 2 \cdot 2 \cdot 4 \cdot \cos(120^\circ) + 16 \cdot 1} \\ &= \sqrt{4 + 2 \cdot 8 \cdot (-\frac{1}{2}) + 16} \\ &= \sqrt{20 - 8} \\ &= \sqrt{12}. \end{aligned}$$

$$\begin{aligned} \|\vec{b}\| &= \sqrt{\langle \vec{b}, \vec{b} \rangle} = \sqrt{\langle \vec{u} - \vec{m}, \vec{u} - \vec{m} \rangle} = \sqrt{\langle \vec{u}, \vec{u} - \vec{m} \rangle - \langle \vec{m}, \vec{u} - \vec{m} \rangle} \\ &= \sqrt{\langle \vec{u}, \vec{u} \rangle - 2\langle \vec{u}, \vec{m} \rangle + \langle \vec{m}, \vec{m} \rangle} = \sqrt{\|\vec{u}\|^2 - 2\langle \vec{u}, \vec{m} \rangle + \|\vec{m}\|^2} = \sqrt{1 + 2 \cdot \frac{1}{2} + 1} = \sqrt{3}. \end{aligned}$$

$$\langle a, b \rangle = \langle 2u + 4v, u - v \rangle$$

$$= \langle 2u, u \rangle - \langle 2u, v \rangle + \langle 4v, u \rangle - \langle 4v, v \rangle$$

$$= \underbrace{2\langle u, u \rangle} - 2\langle u, v \rangle + 4\langle v, u \rangle - 4\langle v, v \rangle$$

$$= 2 \cdot 1 - 2 \cdot \left(-\frac{1}{2}\right) + 4 \cdot \left(-\frac{1}{2}\right) - 4 \cdot 1$$

$$= +2 + 1 - 2 - 4$$

$$= -3$$

$$\langle a, b \rangle = \|a\| \cdot \|b\| \cdot \cos(\widehat{a, b})$$

$$\Rightarrow -3 = \sqrt{2} \cdot \sqrt{3} \cdot \cos(\widehat{a, b})$$

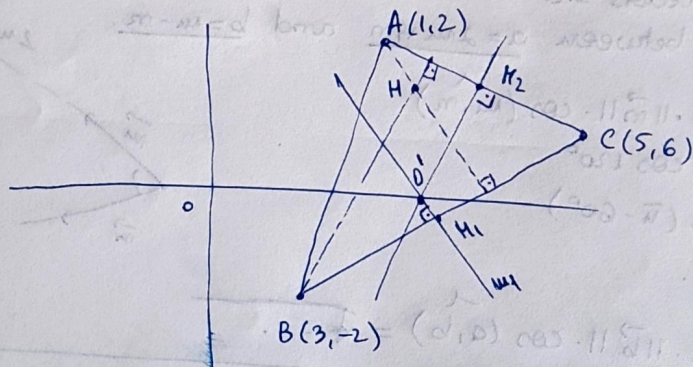
$$\Rightarrow \cos(\widehat{a, b}) = \frac{-3}{\sqrt{2} \cdot \sqrt{3}} = \frac{-3}{2\sqrt{3} \cdot \sqrt{3}} = -\frac{1}{2}$$

$$\Rightarrow \angle(a, b) = 120^\circ$$

3.2. A(1,2), B(3,-2), C(5,6). For $\triangle ABC$:

a) O' ? $\rightarrow$ circumcenter ($\perp$ + mij.)

H? $\rightarrow$ orthocenter ($\perp$).



$O'(x, y)$.

$O' \rightarrow$ n of mediators $\rightarrow$ equations?

let l_1 mediator of BC; $H_1 \rightarrow$ midpoint of BC

l_2 mediator of AC; $H_2 \rightarrow$ midpoint of AC.

$$\vec{OH}_1 = \frac{\vec{OB} + \vec{OC}}{2} \Rightarrow H_1\left(\frac{3+5}{2}, \frac{-2+6}{2}\right) = H_1(4, 2)$$

$$\vec{OH}_2 = \frac{\vec{OA} + \vec{OC}}{2} \Rightarrow H_2\left(\frac{1+5}{2}, \frac{2+6}{2}\right) = H_2(3, 4)$$

let $O'(x, y) \Rightarrow O'H_1 \perp BC$.

$$\Rightarrow \angle(O'H_1, BC) = 0$$

$$\Rightarrow \langle (4-x, 2-y), (5-3, 6-(-2)) \rangle = 0$$

$$(\vec{O'M_1} = M_1 - O' = (4, 2) - (x, y))$$

$$2(4-x) + 8(2-y) = 0$$

$$8 - 2x + 16 - 8y = 0$$

$$-2x - 8y + 24 = 0 \quad | : 2$$

$$-x - 4y + 12 = 0$$

$$\underline{x + 4y = 12.}$$

$$O'M_2 \perp AC$$

$$\Rightarrow \langle \vec{O'M_2}, \vec{AC} \rangle = 0$$

$$\Rightarrow \langle (3-x, 4-y), (4, 4) \rangle = 0.$$

$$4(3-x) + 4(4-y) = 0$$

$$12 - 4x + 16 - 4y = 0$$

$$-x - y + 7 = 0$$

$$\underline{x + y = 7.}$$

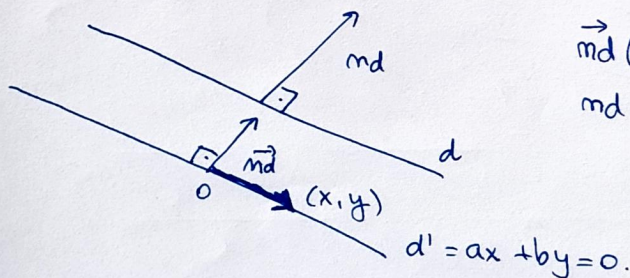
$$\begin{cases} x + 4y = 12 \\ x + y = 7 \end{cases}$$

$$3y = 5 \Rightarrow y = \frac{5}{3}$$

$$x = 7 - \frac{5}{3} = \frac{16}{3}$$

$$\Rightarrow O' \left(\frac{16}{3}, \frac{5}{3} \right)$$

$$d: ax + by + c = 0.$$



$$\vec{md}(a, b)$$

md = normal vector of d .

$$d': ax + by = 0$$

$$\Rightarrow \langle (a, b), (x, y) \rangle = 0$$

$$\Rightarrow (a, b) \perp (x, y).$$

$$m_1: m_1 \in m_1$$

$$\Rightarrow m_1: 2(x-4) + 8(y-3) = 0$$

$$\Rightarrow 2x + 8y - 4 \cdot 2 - 3 \cdot 8 = 0$$

$$\Rightarrow \vec{m}_{m_1} = (2, 8) = \vec{BC} = (2, 8) \Rightarrow \begin{matrix} \alpha = 2 \\ \beta = 8 \end{matrix}$$

normal form of a line.

$$m_1: 2(x-4) + 8(y-3) = 0$$

$$m_1: 2x - 8 + 8y - 24 = 0$$

$$m_1: 2x + 8y - 32 = 0 \quad | :2$$

$$m_1: x + 4y = 16$$

Let $H(x, y)$

$H \rightarrow n$ of $\perp$

$(x, y) \leftarrow$ on the foot.

$(x, y) \in BC$.

$$\vec{AH} \perp \vec{BC}$$

$$\vec{BC}(2, 8)$$

$$m_1: x + 4y = 16$$

$$(\alpha, \beta)(2, 8) = 0$$

$$\Rightarrow 2\alpha + 8\beta = 0 \Rightarrow \alpha + 4\beta = 0.$$

$$\Rightarrow (\alpha, \beta) = (4, -1).$$

$$BC: 4(x-4) - (y-3) = 0.$$

$$BC: y = 4x - 13. \quad \perp \quad m_1: x + 4y = 16$$

?

$$((P, X) - (S, N)) = (0, -1) = \overrightarrow{NM}$$

$$0 = (P-S)8 + (X-N)2$$

$$0 = 48 - 21 + 2X - 8$$

$$5: 0 = 2X + 48 - 29 = 2X + 19$$

$$0 = 2X + 19 - 2X - 19 = 0$$

$$2X = 19 - 19 = 0$$

$$OA \perp SM'O$$

$$O = \langle \overrightarrow{OA}, \overrightarrow{SM'O} \rangle = 0$$

$$O = \langle (M, N), (P-N)(X-S) \rangle = 0$$

$$O = (P-N)N + (X-S)N$$

$$O = PN - 21 + XN - 51$$

$$0 = P + N - X -$$

$$P = N + X$$

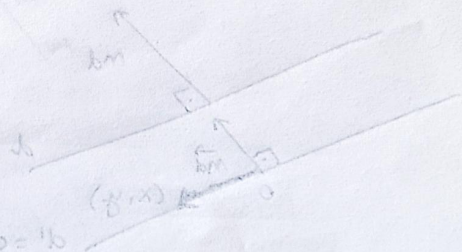
$$\left. \begin{array}{l} 51 = PN + X \\ 4 = P + X \end{array} \right\}$$

$$\frac{P}{5} = \frac{X}{4} \Rightarrow P = \frac{5}{4}X$$

$$\frac{P}{5} = \frac{X}{4} \Rightarrow P = \frac{5}{4}X$$

$$(d, x) \leftarrow$$

to the vector direction = $\vec{bm}$



$$0 = p \cdot d + x \cdot x = 16$$

$$0 = p \cdot d + x \cdot x = 16$$

$$0 = \langle (P, X), (d, x) \rangle = 0$$

$$(P, X) \perp (d, x) \Rightarrow$$

$$m \in m: m$$

$$O = (c-p)q + (p-x)h: m =$$

$$O = q \cdot c - p \cdot q + x \cdot h - p \cdot x =$$

$$8 = q \cdot 5 - 4 \cdot 4 + 3 \cdot 2 - 4 \cdot 4 = 5q - 11 = 8 \Rightarrow 5q = 19 \Rightarrow q = \frac{19}{5}$$

3.3. $A(5,10)$, $B(19,10)$, $C(10,22)$. Incenter of $\triangle ABC$? (3.3)

I = Incenter = point inside $\triangle$ with equal dist. to the three sides.

$AB: y-10=0.$

$\vec{AC} = (10-5, 22-10) = (5,12) \rightarrow$ normal vector: $(12,-5).$

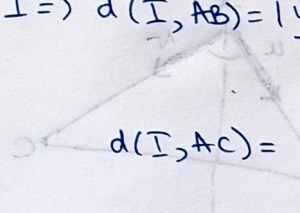
? $AC: 12(x-5) - 5(y-10) = 0. \Rightarrow AC: 12x - 5y - 10 = 0.$

$\vec{BC} = (-9,12) \rightarrow$ normal vector: $(4,3).$

$BC: 4(x-19) + 3(y-10) = 0$

$BC: 4x + 3y - 106 = 0$

$I \Rightarrow d(I, AB) = |y-10| \cdot \frac{y-10}{\text{inside } \triangle}$



$d(I, AC) = \frac{|12x - 5y - 10|}{\sqrt{12^2 + (-5)^2}} = \frac{|12x - 5y - 10|}{13} = \frac{12x - 5y - 10}{13}$

$y-10 = \frac{12x - 5y - 10}{13}$

$\Rightarrow 13y - 130 = 12x - 5y - 10 \Rightarrow 12x - 18y + 120 = 0 \quad | :3$
 $\Rightarrow 4x - 6y + 40 = 0 \quad | :2$
 $\Rightarrow 2x - 3y + 20 = 0.$

$\Rightarrow 3y = 2x + 20.$

$d(I, BC) = \frac{|4x + 3y - 106|}{\sqrt{4^2 + 3^2}} = \frac{|4x + 3y - 106|}{5} = \frac{106 - 4x - 3y}{5}$

$y-10 = \frac{106 - 4x - 3y}{5} \Rightarrow 5y - 50 = 106 - 4x - 3y$

$\Rightarrow 8y = 156 - 4x$

$\Rightarrow x + 2y = 39.$

$\begin{cases} 3y = 2x + 20 \\ x = 39 - 2y \end{cases} \Rightarrow \begin{cases} 3y = 2(39 - 2y) + 20 \\ \Rightarrow 3y = 78 - 4y + 20 \\ \Rightarrow 7y = 98 \end{cases}$

$y = 14$

$x = 39 - 28 = 11.$

$\Rightarrow I(11,14).$

3.4. Cartesian eq. of π if $A(1, -1, 3)$ is orthogonal projection of O onto π .

A is orthogonal projection of O onto $\pi \Rightarrow A \in \pi$

$\vec{OA}$ normal to π .

$$n = \vec{OA} = (1, -1, 3).$$

POINT-NORMAL FORM: $A(1, -1, 3)$:

$$1(x-1) - 1(y+1) + 3(z-3) = 0.$$

$$\Rightarrow x-1-y-1+3z-9=0$$

$$\Rightarrow x-y+3z-11=0.$$

$$\pi: x-y+3z-11=0.$$

3.5. $A(1, 2, -7), B(2, 2, -7), C(3, 4, -5) \Rightarrow \triangle ABC$.

eq. for the angle bisector $\angle CAB$.

$$\vec{AB} = B - A = (1, 0, 0).$$

$$\vec{AC} = C - A = (2, 2, 2) = 2(1, 1, 1).$$

$$|AB| = 1$$

$$|AC| = 2\sqrt{3}.$$

$$u = \frac{\vec{AB}}{|AB|} = \frac{(1, 0, 0)}{1} = (1, 0, 0).$$

$$v = \frac{\vec{AC}}{|AC|} = \frac{2(1, 1, 1)}{2\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right).$$

$$w = u + v = \left(1 + \frac{1}{\sqrt{3}}, 0 + \frac{1}{\sqrt{3}}, 0 + \frac{1}{\sqrt{3}}\right) \cdot \sqrt{3}$$

$$\Rightarrow w = (\sqrt{3}+1, 1, 1).$$

$$\angle(w, u) = 1 + \angle(u, v) = \angle(w, v).$$

$$x-1 = t(\sqrt{3}+1)$$

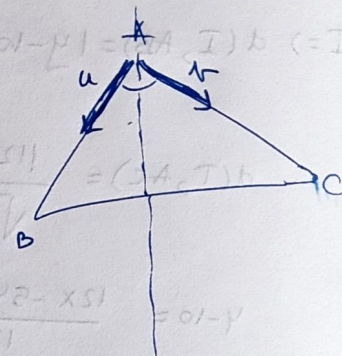
Parametric form: $\begin{cases} x = 1 + t(\sqrt{3}+1) \\ y = 2 + t \cdot 1 \\ z = -7 + t \cdot 1 \end{cases} \Rightarrow \begin{cases} x-1 = t(\sqrt{3}+1) \\ y-2 = t \\ z+7 = t \end{cases}$

Symmetric form: $\frac{x-1}{\sqrt{3}+1} = y-2 = z+7$

Plane of $\triangle ABC$: $\vec{AB} \times \vec{AC} = (0, -2, 2) \sim (0, -1, 1)$

plane ABC:

$$0(x-1) - 1(y-2) + 1(z+7) = 0 \Rightarrow y-z-9=0.$$



3.6) Planes which pass through $P(0,2,0)$, $Q(-1,0,0)$; form 60° angle with the z -axis.

$$\vec{PQ} = (-1, -2, 0).$$

$m = (a, b, c)$ normal vector.

$$m \perp \vec{PQ}$$

$$\Rightarrow -a - 2b = 0 \Rightarrow a = -2b.$$

$$\Rightarrow m(-2b, b, c).$$

z -axis: $k = (0, 0, 1)$.

For a line and a plane, the acute angle is: $\frac{\pi}{2} - \arccos\left(\frac{|k \cdot m|}{|k||m|}\right) = 60^\circ$.

$$\Rightarrow \arccos\left(\frac{|k \cdot m|}{|k||m|}\right) = 30^\circ \quad | \cos$$

$$\Rightarrow \frac{|k \cdot m|}{|k||m|} = \cos 30^\circ = \frac{\sqrt{3}}{2} \quad \Rightarrow \frac{|c|}{\sqrt{(-2b)^2 + b^2 + c^2}} = \frac{\sqrt{3}}{2}$$

$$k = (0, 0, 1)$$

$$\Rightarrow \frac{|c|}{\sqrt{5b^2 + c^2}} = \frac{\sqrt{3}}{2} \quad \Rightarrow \frac{c^2}{5b^2 + c^2} = \frac{3}{4}$$

$$\Rightarrow \frac{c^2}{5b^2 + c^2} = \frac{3}{4}$$

$$\Rightarrow 4c^2 = 15b^2 + 3c^2$$

$$\Rightarrow c^2 = 15b^2$$

$$\Rightarrow c = \pm \sqrt{15}b$$

$$b=1 \Rightarrow m_1(-2, 1, \sqrt{15}), m_2(-2, 1, -\sqrt{15}).$$

$$m_1: \pi_1: -2(x-0) + 1 \cdot (y-2) + \sqrt{15}(z-0) = 0$$

$$P(0,2,0) \Rightarrow \pi_1: -2x + y + \sqrt{15}z - 2 = 0$$

$$m_2: \pi_2: -2(x-0) + (y-2) - \sqrt{15}(z-0) = 0$$

$$\Rightarrow \pi_2: -2x + y - 2 - \sqrt{15}z = 0.$$

3.7. $\triangle ABC$. $|AB|=c$, $|BC|=a$, $|CA|=b$.

$$\langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2} (b^2 + c^2 - a^2)$$

(Law of cosines): $\cos \angle A = \frac{b^2 + c^2 - a^2}{2bc}$

$$\vec{BC} = \vec{AC} - \vec{AB}$$

$$a^2 = |\vec{BC}|^2 = \langle \vec{BC}, \vec{BC} \rangle$$

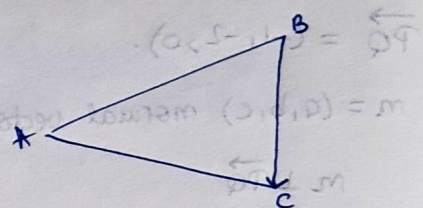
$$a^2 = \langle \vec{AC} - \vec{AB}, \vec{AC} - \vec{AB} \rangle$$

$$a^2 = \langle \vec{AC}, \vec{AC} \rangle - 2 \langle \vec{AC}, \vec{AB} \rangle + \langle \vec{AB}, \vec{AB} \rangle$$

$$a^2 = b^2 - 2 \langle \vec{AC}, \vec{AB} \rangle + c^2$$

$$\Rightarrow 2 \langle \vec{AB}, \vec{AC} \rangle = b^2 + c^2 - a^2$$

$$\Rightarrow \langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2} (b^2 + c^2 - a^2)$$



$$\vec{BA} + \vec{AC} = \vec{BC}$$

$$\vec{BC} = \vec{AC} - \vec{AB}$$

3.8. $\triangle ABC$, $|AB|=c$, $|BC|=a$, $|CA|=b$

$$p = \frac{1}{2} (a + b + c)$$

$$\langle \vec{AB}, \vec{AC} \rangle + bc = 2p(p-a) \quad (1)$$

$$\langle \vec{AB}, \vec{AC} \rangle - bc = -2(p-b)(p-c)$$

$$A_{\triangle ABC} = \sqrt{p(p-a)(p-b)(p-c)}$$

$$\text{Let } \langle \vec{AB}, \vec{AC} \rangle = x = \frac{1}{2} (b^2 + c^2 - a^2)$$

$$(1): \frac{1}{2} (b^2 + c^2 - a^2) + bc = \frac{1}{2} (b^2 + c^2 - a^2 + 2bc) =$$

$$= \frac{1}{2} ((b+c)^2 - a^2) = \frac{1}{2} (\underbrace{(b+c-a)}_{2(p-a)} \underbrace{(b+c+a)}_{2p}) = \frac{1}{2} \cdot 2(p-a) \cdot 2p = 2p(p-a)$$

$$2p = a + b + c \quad | -2a$$

$$2(p-a) = b + c - a$$

$$(2): x - bc = \frac{1}{2} (b^2 + c^2 - a^2) - bc = \frac{1}{2} (b^2 + c^2 - a^2 - 2bc) =$$

$$= \frac{1}{2} ((b-c)^2 - a^2) = \frac{1}{2} (\underbrace{(b-c-a)}_{-2(p-b)} \underbrace{(b-c+a)}_{2(p-c)})$$

$$= \frac{1}{2} \cdot (-2(p-b)) \cdot 2(p-c) = -2(p-b)(p-c)$$

$$b - c - a = -(a - b + c) = -2(p-b)$$

$$b - c + a = -(-a - b + c) = 2(p-c)$$

$$A_{[ABC]} = \frac{1}{2} \cdot |AB| \cdot |AC| \cdot \sin A = \frac{1}{2} bc \cdot \sin A.$$

$$\Rightarrow 2 \cdot A = bc \cdot \sin A \quad |^2$$

$$\Rightarrow 4A^2 = b^2 c^2 \sin^2 A$$

$$\cos A = \frac{x}{bc} \quad |^2 \Rightarrow \cos^2 A = \frac{x^2}{b^2 c^2}$$

$$4A^2 = b^2 c^2 (1 - \cos^2 A) = b^2 c^2 - b^2 c^2 \cdot \cos^2 A = b^2 c^2 - x^2 = (bc - x)(bc + x)$$

$$4A^2 = (+2(p-b)(p-c))(2p(p-a)) =$$

$$4A^2 = +4p(p-a)(p-b)(p-c)$$

$$A^2 = p(p-a)(p-b)(p-c).$$